

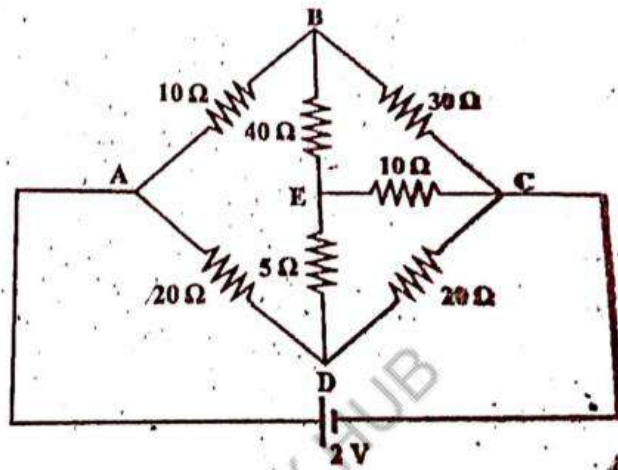
B.E. I Yr. (Common for ETC-A, ETC-B)
Mid Test-I (Date – 02/12/2022)
Sub: Electrical Engineering (EIR2C4)

Time: 70 min

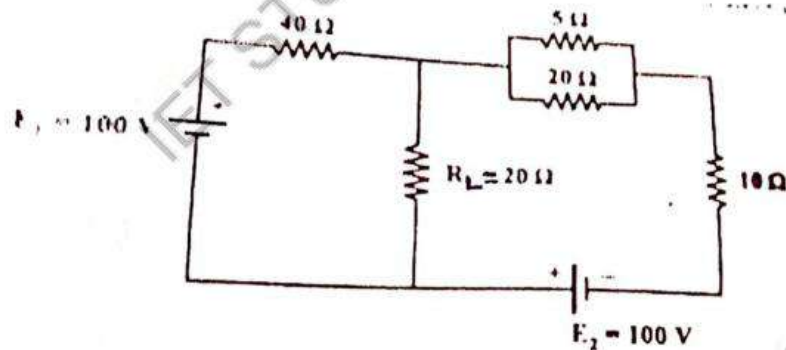
Mark: 20

Note: All questions carry equal marks.

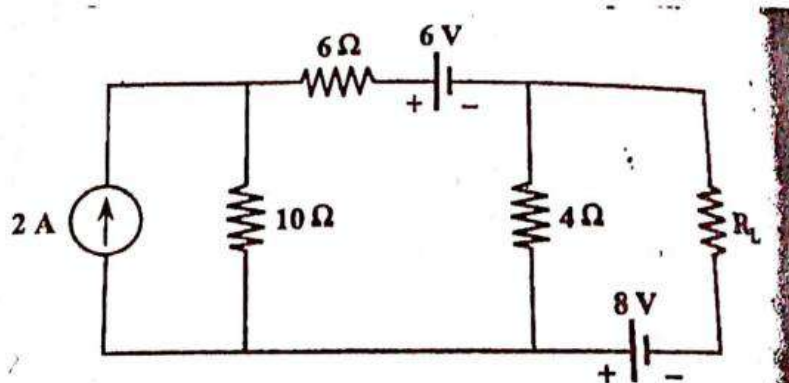
1. An alternating current of frequency 50Hz has a maximum value of 100A. Calculate (a) its value $1/600$ second after the instant the current is zero and its value decreasing thereafter (b) how many seconds after the instant the current is zero (increasing thereafter) will the current attain the value of 86.6 A.
2. Find out the value of current in branch BE in the network shown below:



3. Find current in 20ohm resistor using superposition theorem.



4. Find the value of R_L for which source will transfer maximum power to the load. Also find maximum power transferred.



B.E. I Yr. (Common for IT A, IT B and Civil)
Mid Test-I (Date - 13/03/2024)
Sub: Electrical Engineering (2EIRC4)

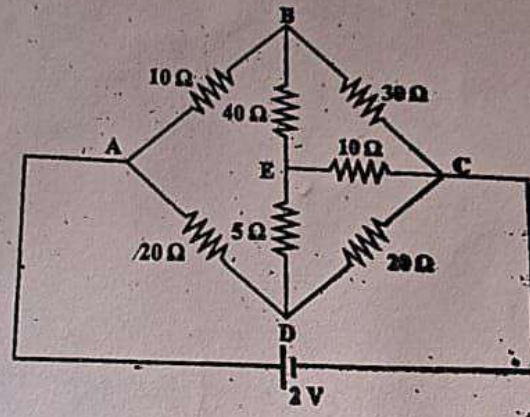
M311-35

Time: 70 min

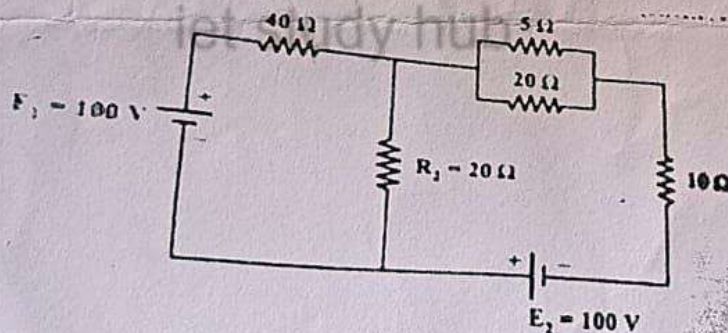
Mark: 20

Note: All questions carry equal marks.

1. An alternating current varying sinusoidally with a frequency of 50 Hz has an RMS value of 20 A. write down the equation for the instantaneous value and find this values at (1) 0.0025 seconds (2) 0.0125 seconds after passing through a positive maximum value, At what time measured from a positive maximum value, will the instantaneous current be 14.14 A.
2. Find out the value of current in branch BE in the network shown below:



3. Find current in 20ohm resistor using superposition theorem.



4. Find the value of R_L for which source will transfer maximum power to the load. Also find maximum power transferred.

